MIL-C-746A(MU)
10 October 1968
SUPERSEDING
JAN-C-746 (In Part)
30 March 1949
(See 6.2)

# MILITARY SPECIFICATION
# CARTRIDGE, GRENADE: RIFLE, CALIBER .30, M3

## 1. SCOPE

1.1 This specification covers Cartridge, Grenade: Rifle, Caliber .30, M3 intended for propelling grenades from a launcher attached to Rifle, U.S., Caliber .30, M1.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-636 | - Visual Inspection Standards for Small Arms Ammunition through Caliber .50 |
| MIL-STD-644 | - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition |

DRAWINGS

U.S. Army Munitions Command

| | |
|---|---|
| C6173949 | - Cartridge, Grenade, Rifle, Caliber .30, M3 |
| C7643674 | - Classification of Cartridge Case Defects |
| IEL-6173949 | - Inspection Equipment List for Cartridge, Grenade, Rifle, Caliber .30, M3 |

FSC 1330

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2.1 (Cont'd)

PUBLICATIONS

TECP 700-700, Vol. III - Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurment functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on drawing C6173949, referenced specifications and the following:

3.2 Velocity.- The sample cartridge which has been conditioned at 68° to 72° Fahrenheit (F) shall impart a velocity of 153 feet per second (ft/sec) plus or minus 5 ft/sec to rifle grenades weighing 1.56 ± .01 pounds. The velocity shall be measured at 5 feet 6 inches beyond the forward end of the grenade when fully positioned for launching. The standard deviation of velocities shall not exceed 2 ft/sec.

3.3 Function and casualty.- The cartridge shall function without casualty.

3.4 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.4.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.4.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discoloration, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 200 cartridges. The sample shall be manufactured using the same materials, equipment, process and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.



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4.3.2.1 (Cont'd)

TABLE I

| No. | Defect and Method of Inspection | Critical | Major | Minor | Major or Minor |
|---|---|---|---|---|---|
| | Visual 1/ | | | | |
| | Cartridge | | | | |
| 1 | Discolored, dirty, oily, smeared | | | X | |
| 2 | Corroded or stained, if etched | | X | | |
| | Case | | | | |
| 4 | Round head | | X | | |
| 5 | Dent | | | | X |
| 6 | Split case | X | | | |
| 7 | Perforated case | X | | | |
| 8 | Draw scratch | | | | X |
| 9 | Scratch | | | X | |
| 10 | Beveled underside of head | | X | | |
| 12 | Scaly metal | | | | X |
| 13 | No chamfer on head (rim) | | X | | |
| 14 | Fold | | | X | |
| 15 | Wrinkle | | | X | |
| 17 | Bulge | | | X | |
| 18 | Illegible or missing head stamp | | | X | |
| 19 | Defective head | | | X | |
| | Primer | | | | |
| 32 | No primer | X | | | |
| 33 | Cocked primer | X | | | |
| 34 | Inverted primer | X | | | |
| 35 | Loose primer | | X | | |
| 36 | Nicked or dented primer | | | X | |
| 37 | No waterproofing material (primer pocket joint) | | | X | |
| 38 | Defective crimp | | | X | |
| | Mouth | | | | |
| 39 | Missing wad | X | | | |
| 40 | Wad lacquer missing or inadequate | | X | | |
| 41 | Wad cannelure missing or incorrect | | X | | |
| 42 | Mouth crimp missing or incorrect | | | X | |
| 43 | Opening between wad and case wall | | X | | |



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4.3.2.1 (Cont'd)

TABLE I (Cont'd)

| No. | Defect and Method of Inspection | Critical | Major | Minor | Major or Minor |
|---|---|---|---|---|---|
| | Gaging | | | | |
| 44 | Cartridge profile failure (requiring more than 20 lbs. dead weight to insert in profile and alignment gage) | | X | | |
| 45 | Diameter of extractor groove | | | X | |
| 46 | Diameter of head | | X | | |
| 47 | Thickness of head, max. | | | X | |
| 48 | Thickness of head, min. | | X | | |
| 49 | Depth of primer | | X | | |
| 50 | Length to shoulder | | X | | |
| | Weighing | | | | |
| 51 | Weight, min. | X | | | |

1/ Refer to MIL-STD-636 for visual standards for defects 1 through 38. While no photographs of grenade cartridges are included in this standard, the cartridge case defects illustrated in the Caliber .30 section can be used. In the event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

| Test | Number of Cartridges | Requirement Paragraph |
|---|---|---|
| Velocity 1/ | 25 | 3.2 |
| Function and casualty | -- | 3.3 |



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4.3.3.1 (Cont'd)

1/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

2/ Observation for function and casualty shall be made during firing of the velocity test. The lot shall be rejected when firing defects exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantity of cartridges specified for the velocity test shall be fired. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

TABLE III

| Defects | Acceptance Number: First Sample | Acceptance Number: Cumulative (First & Second Sample) |
|---|---|---|
| 1. Misfire | 0 | 1 |
| 2. Primer leak | | |
| a. Perforation in firing pin indent in primer cup | 1 | 2 |
| b. Escape of gas through primer cup other than 2a. | 0 | 2 |
| c. Escape of gas around primer cup more than 50% of periphery | 1 | 2 |
| d. Blown primer or primer falls out of pocket on retraction of bolt | 0 | 1 |
| e. Primer remains in pocket but is physically loose | 0 | 2 |



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4.3.3.2 (Cont'd)

TABLE III (Cont'd)

| Defects | Acceptance Number First Sample | Acceptance Number Cumulative (First & Second Sample) |
|---|---|---|
| 3. Case casualties 1/ | | |
| a. Longitudinal split 2/ | | |
| (1) Body (J) | 0 | 2 |
| (2) Body (K) | 0 | 1 |
| (3) To head (L) | 0 | 1 |
| (4) Through head (M) | 0 | 1 |
| b. Circumferential rupture 2/ | | |
| (1) Partial, body (J) | 0 | 2 |
| (2) Partial, body (K) | 0 | 1 |
| (3) Partial, head (L) | 0 | 1 |
| (4) Complete | 0 | 1 |
| 4. Failure to extract | 0 | 1 |

1/ No penalty shall be charged against a fired case with mouth partially or completely burned. If, however, such fired cartridge case shows the occurrence of some other firing casualty, exclusive of failure of rosebud crimp to open, such other casualty shall be penalized in accordance with Table III.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100 percent examination of the cartridge shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a dummy, blank or any bullet type cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-6173949.



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4.4 Test methods and procedures.

4.4.1 Velocity.- The test shall be conducted in accordance with Publication TECP 700-700, Vol. III, using equipment listed on IEL-6173949. Proof-slug(s), weighing 1.56 ± .01 pounds each, shall be used in lieu of a practice grenade(s).

4.4.2 Function and casualty.- Observation for firing defects listed in Table III shall be made during the firing of the velocity test. Retest shall be fired in accordance with the procedure prescribed for the velocity test in Publication TECP 700-700, Vol. III, except that all references pertaining to obtaining of velocity data shall be disregarded.

4.4.3 Defect penalty.- If, in the velocity test, the occurrence of a firing defect prevents the obtaining of a reliable result, an additional shot shall be fired. The velocity test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level A (Worldwide shipment).- The cartridges shall be packed as required by the contracting officer.

5.2 Marking and labeling.- Marking for shipment shall be as specified by the contract.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provisions for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance of inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.



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6.2 Supersession data.- This specification together with MIL-C-60839(MU), Cartridge, Grenade, Carbine, Caliber .30, M6 supersedes JAN-C-746 dated 30 March 1949. Cartridge, Grenade, Auxiliary, M7 has been reclassified to obsolete status.

Custodian:

Army - MU

Preparing activity:

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